



P.I.G.D.TM

Plant Image and Growth Detector

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Sponsored by: MarsFarm

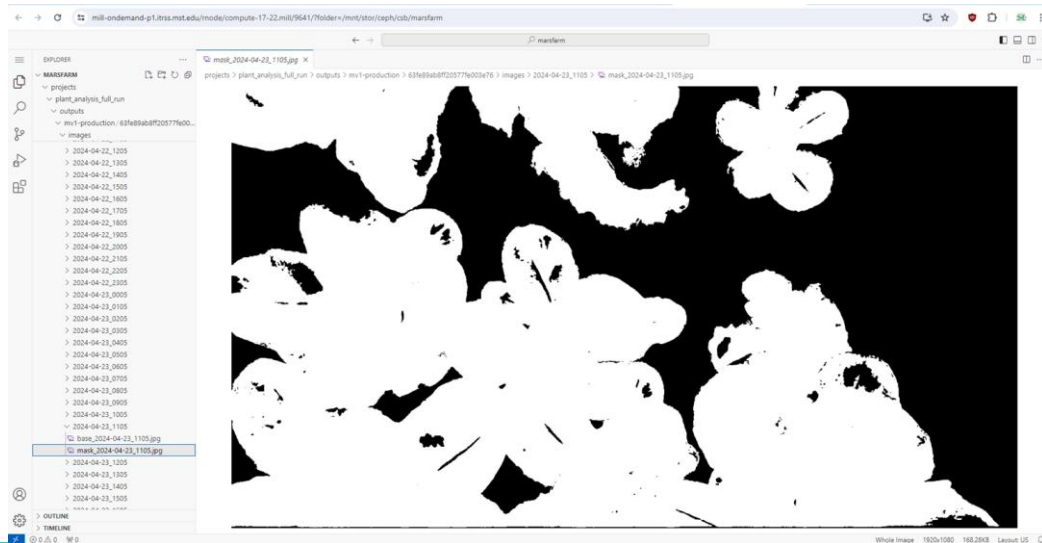
Goal

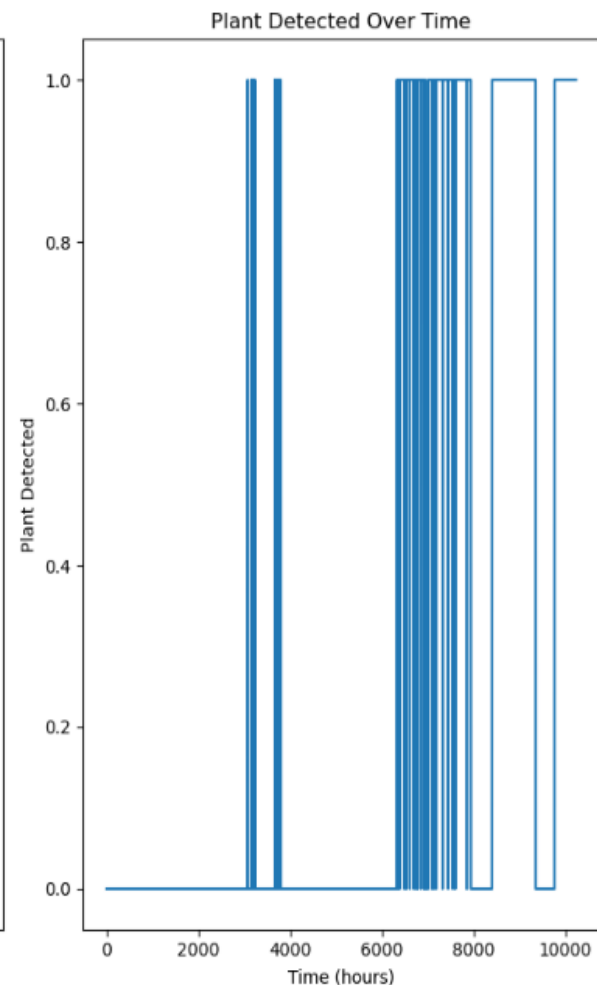
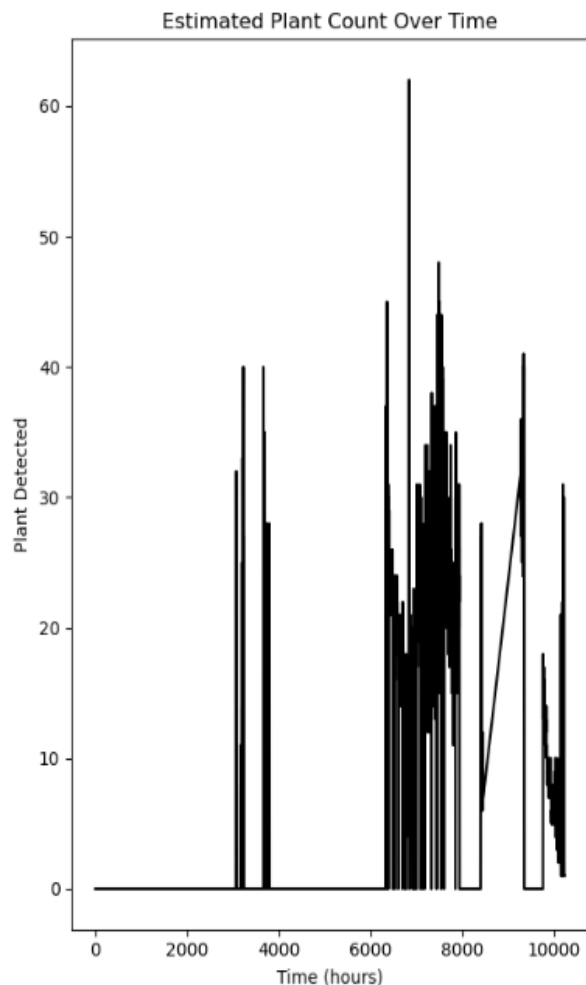
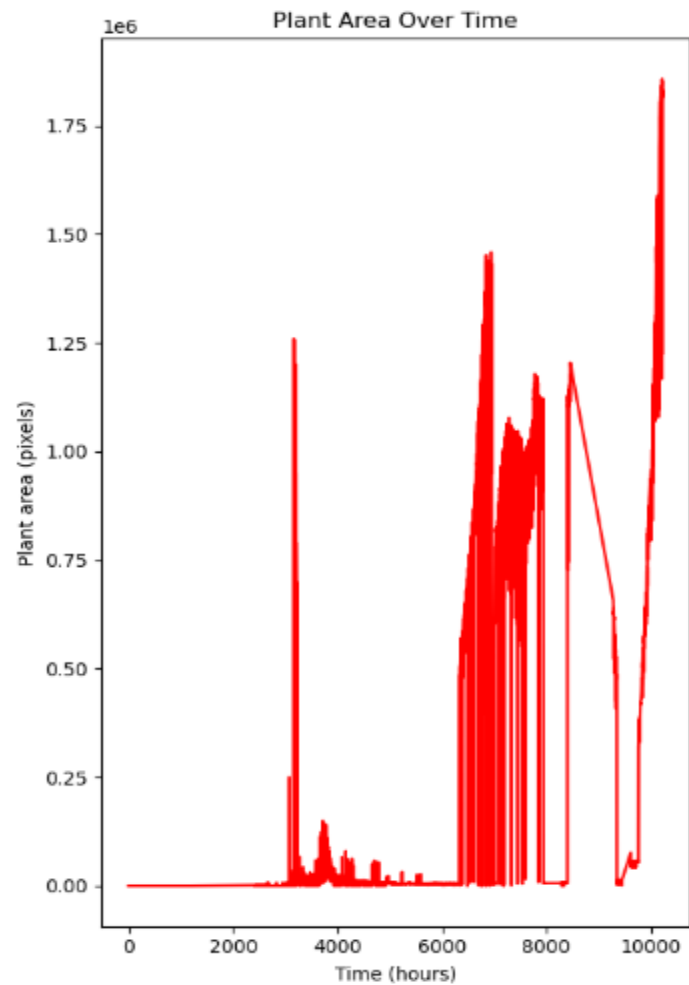
Can we build an AI-powered program to detect and measure healthy plant growth based on pictures?



The Future Is Now!

- Snap some pictures (or a few thousand)
- Upload to the **P.I.G.D™** interface
- We will tell you:
 - If you are looking at a plant
 - How many (alive) plants there are
 - The area of the plants in pixels
 - And more!





Video

Home - Google Drive | Resources / Help Guide - Google | My Interactive Sessions - RStudio | Stream | Microsoft 365 | New Recording - Camera

milli-ondemand-p1.frss.mtu.edu/pun/syn/dashboard/batch_connect/sessions

App* Files* Jobs* Clusters* Interactive Apps* My Interactive Sessions

Help Logged in as srikSev Log Out

Session was successfully created.

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Interactive Apps

- Desktops
 - Mill Desktop
 - GLX
 - ANSYS Workbench
 - Comsol Multiphysics
 - Matlab
- Servers
 - Vscode
 - Jupyter Notebook
 - RStudio Server

Vscode (112952) 1 node | 1 core | Running

Host: [D_compute-12-13.mill](#) Delete

Created at: 2024-05-01 12:22:33 CDT

Time Remaining: 55 minutes

Session ID: e3e8dbf7-2efb-4bcd-b9eb-819af265d159

Connect to VS Code

Vscode (112219) Completed | [?] | [x]

Created at: 2024-04-30 14:13:40 CDT Delete

Session ID: 2f60279b-ca83-485d-a599-763e325394b3

For debugging purposes, this card will be retained for 6 more days.

powered by **OPEN OnDemand**

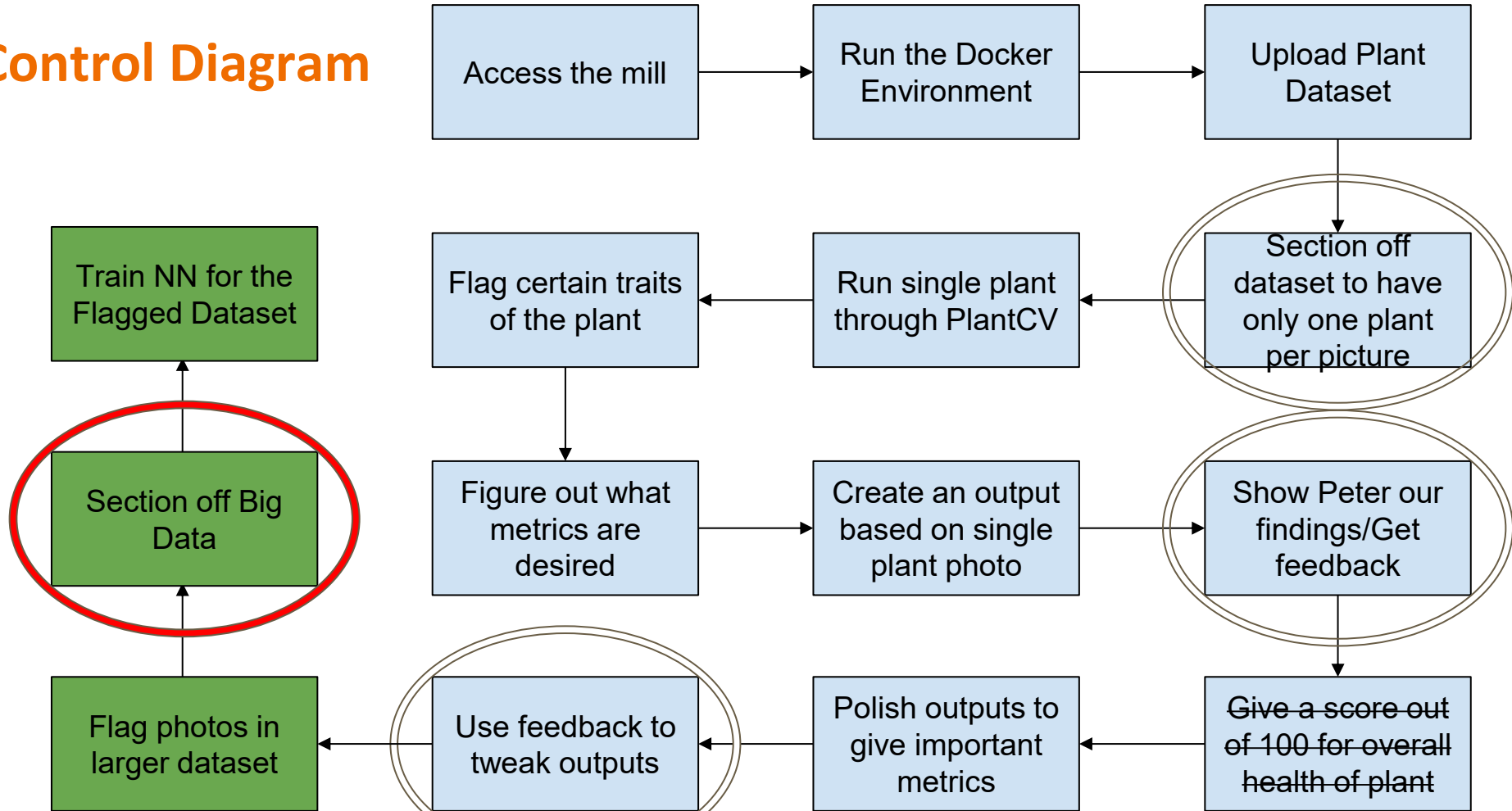
maliniasuri-my.sharepoint.com is sharing your screen. Stop sharing Hide

OnDemand version: 3.0.3

Type here to search

68°F Mostly cloudy 12:27 PM 5/1/2024

Control Diagram



Why should you give us more money and time?

- Want to build labelled dataset with citizen science
- Tell people their crop type, if their plant is sick
- Effectively parse through a large dataset and trim the garbage

Citizen Science Project + Photo Dataset = NN we want

Questions?



Appendices

How can a computer have a green thumb



- PlantCV & OpenCV
 - open-source image analysis software
- YOLOv8 & HuggingFace
 - Object detection with Hugging Face more specific on plant detection
 - Tried using but proved to be ineffective
- Starting dataset of a couple thousand Plant photos
 - Purple Basil
 - Tomatoes
 - Peas
- Expanded Data Set mv1_production
 - Approximately 126k Photos with more being added daily
 - Tagging photos with plant or no plant



How the project has changed

- Tested out various masks and a couple of Neural Networks to find an optimal plant mask and try to detect plant species and diseases
- Transitioned from small dataset of about ~1k images locally stored to ~125k images on an S3 bucket
- The masks worked better than intended